H2 Geography Paper 1 Suggested Answers

Section A

Cluster 1: Development, Economy and Environment

1 Resource 1 shows a cartoon depicting inequality at World Trade Organization (WTO) Talks. Resource 2 shows World Bank lending by region in 2021. Resource 3 shows the WTO membership and global trade share (1995 – 2020).

(a) Suggest reasons for the growing interdependence between places in the global economy. [4]

Award 1 mark for each explanation for the growing interdependence between places in the global economy, to a maximum of 4 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

1. Expansion of Global Trade Networks [1 mark]: The liberalisation of trade policies and reduction of trade barriers (e.g. tariffs, quotas) through multilateral agreements (e.g. WTO) have enabled countries to exchange goods and services more freely, leading to increased reliance on imports and exports across regions [1 mark].
2. Rise of Transnational Corporations (TNCs) [1 mark]: TNCs have developed global production networks (GPNs) where production processes are distributed across multiple countries (e.g. raw materials sourced from one country, manufacturing in another, and consumption elsewhere), creating interdependence among countries involved in different production stages [1 mark].
3. Increased Flow of Capital and Investment [1 mark]: Global financial markets have become more integrated, with Foreign Direct Investment (FDI) and portfolio investments flowing across borders. Developing economies, in particular, rely on capital inflows from developed countries and multilateral institutions to finance infrastructure and growth [1 mark].
4. Labour Migration and Remittances [1 mark]: Labour mobility across borders has led to a flow of skilled and unskilled workers between countries. Remittances sent back home by migrant workers contribute significantly to the GDP of many developing nations, creating economic ties between sending and receiving countries [1 mark].

(b) Describe the criticisms shown in Resource 1 on the role of WTO in the global economy. [4]

Award 1 mark for each description of criticisms on the role of WTO, to a maximum of 4 marks.
Award a maximum of 1 additional mark for further development of each description, where applicable.

- WTO decision-making dominated by powerful actors [1 mark].: The cartoon suggests that the WTO, while intended as a multilateral platform, is heavily influenced by the governments of rich nations and large multinational corporations. This creates a power imbalance in shaping global trade rules. [1 mark].
- Limited influence of poorer members in WTO negotiations [1 mark].: The ants symbolise developing countries that are sidelined in WTO talks, highlighting the criticism that poorer nations often have little say in trade negotiations that directly affect their economies. [1 mark].
- WTO trade rules can disadvantage weaker economies [1 mark].: The caption, “When elephants mate, many ants get crushed,” reflects how WTO-led liberalisation, such as reductions in subsidies or tariff barriers, may benefit stronger economies but place smaller, less competitive economies at risk of harm. [1 mark].

- WTO criticised for inequitable outcomes in global trade [1 mark].: The cartoon questions the WTO's role as a supposedly neutral regulator of trade. In practice, its processes often amplify the voices of powerful economies, undermining fairness and inclusiveness in global economic governance. [1 mark].

(c) Using Resource 2, describe the pattern of World Bank lending in 2021. [5]

Award 1 mark for each description of the pattern of World Bank lending, to a maximum of 5 marks.
Award a maximum of 1 additional mark for further development of each description, where applicable.

- The spatial distribution shows a clear development-oriented pattern, with greater assistance to less developed or economically vulnerable regions [1 mark].
- Sub-Saharan Africa (US$29.4 billion) and South Asia (US$21.8 billion) received the highest amounts of lending, indicating prioritisation of support for lower-income regions [1 mark].
- Europe & Central Asia (US$11.9 billion) and Middle East & North Africa (US$7.6 billion) received the lowest amounts, aligning with its development mandate [1 mark].
- The lending to Sub-Saharan Africa (US$29.4 billion) exceeds that of the next highest region, South Asia, by a notable margin of US$7.6 billion [1 mark].
- Sub-Saharan Africa received nearly four times more than the Middle East & North Africa, highlighting its distinct position in lending patterns [1 mark].

(d) With reference to Resource 3, explain why WTO membership and its global trade share have changed from 1995 to 2020. [6]

Award 1 mark for each explanation on why WTO membership and its global trade share have changed from 1885 to 2020, to a maximum of 6 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Reasons for the changes of WTO membership and its global trade share from 1995 to 2020:

1. Trade liberalisation encouraged more countries to join the WTO
   Membership increased from 128 in 1995 to 164 in 2020 because the WTO offers reduced tariffs, removal of quotas, and elimination of discriminatory trade practices [1 mark]. This made it attractive for countries seeking greater access to global markets, leading to a broader membership base and more trade conducted within the WTO framework [1 mark].
2. WTO provides a rules-based system for trade governance
   Countries joined to benefit from the WTO's dispute settlement mechanism and its platform for multilateral negotiations, which provide stability, predictability, and fairness in market access [1 mark]. This boosted trade among members as businesses could operate with more certainty about trade regulations and enforcement [1 mark].
3. Expansion of South–South trade among WTO members
   Beyond traditional North–South trade, developing countries began trading more with each other due to lower barriers under WTO agreements [1 mark]. This increased the volume of trade within the WTO membership, raising the members' share of global trade from 90% in 1995 to 97% in 2020 [1 mark].
4. Growth of Global Production Networks (GPNs)
   TNCs expanded GPNs, distributing different stages of production (e.g. sourcing, assembly, distribution) across multiple WTO member states [1 mark]. This interlinkage amplified trade flows between members, as intermediate goods and services moved repeatedly across borders before reaching final consumers [1 mark].

(e) Using Resources 2 and 3, explain how multilateral institutions can promote economic development. [5]

Award 1 mark for each explanation on how multilateral institutions can promote economic development, to a maximum of 5 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Ways in which multilateral institutions can promote economic development:

1. By providing targeted financial assistance to less developed regions [1 mark]
   Resource 2 shows that Sub-Saharan Africa received the highest World Bank lending (US$29.4 billion), followed by South Asia (US$21.8 billion), reflecting the World Bank's role in directing capital towards regions with greater development needs, such as poverty alleviation, infrastructure, education, and healthcare [1 mark].
2. By improving access to global markets for developing countries [1 mark]
   As shown in Resource 3, WTO membership expanded from 128 in 1995 to 164 in 2020, allowing more countries—especially developing and transition economies—to integrate into global trade, which supports export-led growth and industrialisation [1 mark].
3. By encouraging inclusive economic participation and integration [1 mark]
   The increase in global trade share of WTO members (from 90% in 1995 to 97% in 2020) shown in Resource 3 suggests greater participation of developing countries in the global economy, leading to expanded trade opportunities, job creation, and income growth [1 mark].

(f) Explain why multilateral institutions may be more influential than other actors in shaping the global economy. [6]

Award 1 mark for each explanation on why multilateral institutions may be more influential than other actors in shaping the global economy, to a maximum of 6 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Reason for multilateral institutions more influential than other actors:

1. They set global trade and development rules that bind multiple countries [1 mark]
   Institutions like the WTO establish legal frameworks for trade, such as the Most-Favoured-Nation (MFN) principle and dispute resolution systems, which shape how member states and corporations operate within the global economy [1 mark].
2. They facilitate inclusion of developing countries into the global economy [1 mark]
   As seen in Resource 3, WTO membership expanded from 128 in 1995 to 164 in 2020, with developing countries accounting for a growing share of global trade (from 90% to 97%). This demonstrates their role in integrating more economies into the global system [1 mark].
3. They provide financial capital and technical expertise for development [1 mark]
   Resource 2 shows how the World Bank channels significant development lending to less developed regions such as Sub-Saharan Africa (US$29.4 billion) and South Asia (US$21.8 billion), funding essential infrastructure, healthcare, and education, which individual firms or labour actors cannot provide [1 mark].

Section B

Cluster 2: Tropical Environments

2 Resource 4 shows the position of the Intertropical Convergence Zone (ITCZ) and wind prevailing wind directions in January and July. Resource 5 shows a climate graph of Kolkata, India. Resource 6 shows a satellite image of Ganges Delta with its major rivers. Resource 7 shows a section of Padma River in the Ganges Delta.

(a) With reference to Resource 4, describe the differences in the features of the ITCZ and wind systems in January and July. [4]

Award 1 mark for each description on the differences in the features of the ITCZ and wind systems in January and July, to a maximum of 4 marks.
Award a maximum of 1 additional mark for further development of each description, where applicable.

- In January, the ITCZ is located south of the equator, roughly over the southern Indian Ocean, while in July, the ITCZ shifts northwards, extending over the Indian subcontinent, especially northern India and the Bay of Bengal region [1 mark].
- The January ITCZ shows less fluctuations over the ocean, in contrast July ITCZ has greater fluctuations over the landmass. [1 mark].
- The wind systems in January are dominated by the Northeast Monsoon, where dry, cool winds blow from the northeast to the southwest, originating from the Asian landmass and flowing toward the equator [1 mark].
- In contrast, during July, the prevailing winds are part of the Southwest Monsoon, which brings moist, warm air from the southwest, having picked up moisture from the Arabian Sea and Indian Ocean [1 mark].

(b) With reference to Resources 4, account for the climatic characteristics of Kolkata, India as shown in Resource 5. [6]

Award 1 mark for each explanation on the climatic characteristics of Kolkata, to a maximum of 6 marks. Maximum 3 marks for either rainfall or temperature only.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

- The wet season occurs from June to September, with the highest rainfall in July (approx. 370 mm), due to the northward shift of the ITCZ and the arrival of the Southwest Monsoon winds bringing moisture-laden air from the Indian Ocean (Resource 4) [1 mark].
- The dry season spans from November to April, where rainfall drops below 25 mm per month, with the driest month being December (~10 mm), explained by the southward displacement of the ITCZ and dominance of dry Northeast Monsoon winds from the continental interior (Resource 4) [1 mark].
- Total annual rainfall is 1656 mm, indicating a seasonal concentration of precipitation, with over 75% of rainfall falling during the wet season [1 mark].
- Kolkata's temperature remains high and relatively stable year-round, with monthly averages ranging from about 19°C in January to nearly 30°C in May, supporting its classification as a tropical climate [1 mark].
- Highest temperature in May (30°C) due to absence or little cloud cover just before the wet season and the position of the overhead sun (ITCZ) – high sun at its location [1 mark].
- Lowest temperature in January (19°C) due to low sun as ITCZ (position of overhead sun) is away from its location [1 mark].

(c) Describe the characteristics of the Ganges delta as shown in Resource 6. [4]

Award 1 mark for each description on the Ganges delta characteristics, to a maximum of 4 marks.

Award a maximum of 1 additional mark for further development of each description, where applicable.

- The Ganges Delta has a distinct triangular (or arcuate) shape, with its apex near the confluence of the Ganges and Brahmaputra rivers and its base spreading along the Bay of Bengal [1 mark].
- The delta is formed by major rivers, including the Ganges (or Padma in lower stretches), Brahmaputra, and Meghna, which branch into multiple distributaries across the deltaic plain [1 mark].
- The delta appears densely dissected by a network of river channels and distributaries, indicating active sediment deposition and fluvial reworking [1 mark].
- It spans a transboundary region, covering eastern India (e.g., Kolkata) and much of Bangladesh (e.g., Dhaka), showing that it is one of the largest and most extensive deltas in the world [1 mark].

(d) Explain the impacts of human activities on ecosystem services provided by deltas. [6]

Award 1 mark for each explanation on the impacts of human activities on ecosystem services provided by deltas, to a maximum of 6 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

- Upstream dam construction (e.g., on the Ganges or Brahmaputra) reduces sediment load reaching the delta, which in turn disrupts natural nutrient replenishment, degrading provisioning services such as fertile soils for agriculture [1 mark].
- Agricultural intensification on deltaic floodplains involves the use of chemical fertilisers and pesticides, leading to pollution of surface and groundwater, which undermines regulating services like water purification [1 mark].
- Urbanisation and land reclamation for housing and infrastructure reduce wetlands and mangroves, impairing regulating services such as flood mitigation, groundwater recharge, and carbon sequestration [1 mark].
- Aquaculture expansion, especially intensive shrimp farming, causes salinisation of soils and groundwater due to brackish water intrusion, damaging supporting services like soil formation and nutrient cycling [1 mark].
- Overfishing and habitat destruction reduce the availability of provisioning services such as fish stocks and forest products (e.g., fuelwood and medicinal plants from mangroves) [1 mark].
- Channelisation and embankment construction along riverbanks prevent regular flooding of the deltaic plain, disrupting the natural flood pulse, which is vital for maintaining biodiversity and supporting cultural services such as traditional livelihoods and spiritual practices tied to seasonal floods [1 mark].

(e) Describe the characteristics of the fluvial landform as shown in Resource 7. [4]

Award 1 mark for each description on fluvial landform characteristics, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

- The landform is a braided river, characterised by multiple interweaving channels that split and rejoin around elongated mid-channel bars or islands, known locally as chars [1 mark].
- The channel is wide and shallow (high width-to-depth ratio), with evidence of extensive sediment deposition (mostly made up coarse-grained e.g. sand or gravel), as seen in the exposed sandy bars within the channel [1 mark].
- The channel is straight having sinuosity index of less than 1.5 [1 mark].

- Some mid-channel bars are colonized by vegetation signifying stable conditions whilst those without are unstable and frequent re-working of sediments (changing their shapes) [1 mark].

(f) Explain the fluvial processes that led to the formation of the landform shown in Resource 7. [6]

Award 1 mark for each explanation on the fluvial processes leading to the formation of the landform, to a maximum of 6 marks.
Award a maximum of 1 additional mark for further development of each explanation, where applicable.

- The Padma River carries a high sediment load, primarily from the Ganges and Brahmaputra headwaters in the Himalayas, which promotes extensive deposition within the channel during periods of reduced flow [1 mark].
- Lateral erosion during high discharge causes the river to widen and shift its course across the floodplain, contributing to bank instability and the frequent reworking of sediments [1 mark].
- During the monsoon season, high-energy flows lead to vertical and lateral erosion, but as the flow velocity decreases downstream or between floods, the sediment is deposited, forming mid-channel bars [1 mark].
- During their formation, these bars emerge above the water level, splitting the river into multiple interconnected channels, a process typical in rivers with a high sediment-to-discharge ratio [1 mark].
- The deposited bars are continually reworked by shifting flows, resulting in a dynamic channel system where the position and number of channels change over time—a hallmark of braided rivers [1 mark].
- Seasonal fluctuations in discharge, with dramatic differences between dry season base flow and peak monsoon flood, accelerate the alternation between erosion and deposition, maintaining the braided pattern [1 mark].

Section B

Cluster 1: Development Economy and Environment

3 ‘It is not possible to solve conflicts over transboundary sources of water supply.’

Evaluate this statement. [20]

Possible Approaches:
Candidates should approach this question by evaluating the extent to which conflicts over transboundary water sources can be resolved, using a range of case studies from different spatial and political contexts. They should assess the effectiveness of strategies such as treaties, institutional frameworks, and multilateral cooperation in resolving or mitigating conflicts, and whether these efforts have led to long-term, equitable, and sustainable outcomes.

Responses may compare the success of different strategies based on criteria such as their durability over time, ability to address root causes (e.g., power asymmetries, water scarcity), involvement of external mediators or international organisations, and the influence of contextual factors like climate variability, political will, and access to alternative water sources.

Candidates should also consider whether full resolution is possible, or if only partial mitigation can be achieved. Stronger responses will explain why certain regions succeed more than others by applying geographical perspectives such as scale, place, time, and systems thinking.

Levels marked using Generic Level Descriptors for 20m H2 essays

4 ‘Global Production Networks (GPNs) of transnational corporations contribute to sustainable development in countries.’

Evaluate this statement. [20]

Possible Approaches
Candidates should evaluate the extent to which Global Production Networks (GPNs) contribute to sustainable development by considering their economic, social, and environmental impacts. Responses should examine both positive effects—such as job creation, infrastructure, and knowledge transfer—and negative consequences like labour exploitation, environmental harm, and uneven development.

A balanced approach may weigh how GPNs support or hinder sustainability, noting variations across contexts. For example, strong governance in countries like Singapore or Vietnam can align GPNs with sustainability goals, while weak institutions in places like the DRC may worsen negative outcomes. The role of states and multilateral institutions in shaping these outcomes should also be considered.

Stronger responses will use relevant case studies, apply geographical concepts such as scale and place, and offer a clear, reasoned judgement that reflects the complexity of GPNs and their impacts.

Levels marked using Generic Level Descriptors for 20m H2 essays

Cluster 2: Tropical Environments

5 Evaluate the significance of the El Niño–Southern Oscillation (ENSO) in affecting rainfall variability in tropical regions. [20]

Possible Approaches

Candidates may highlight ENSO's significance in regions influenced by the Pacific Ocean. During El Niño, convection shifts eastwards, bringing drought to Indonesia and northern Australia and floods to Peru and Ecuador, while La Niña reverses these effects, intensifying rainfall in Papua New Guinea and Queensland. ENSO also affects South Asia by weakening or strengthening the Indian monsoon.

Alternatively, candidates may stress the Hadley circulation's overarching control of tropical rainfall. The ITCZ generates continuous rainfall in rainforest (Af) climates, seasonal rains in savannah (Aw) regions, and brief wet seasons in steppe (BSh) margins, while the STHP maintains persistent aridity in deserts (BWh). Compared to ENSO's episodic influence, these are permanent and predictable controls.

Candidates might also note the localised impact of tropical cyclones, which bring intense short-term rainfall to coastal tropics. Stronger responses will evaluate relative importance, concluding that while ENSO drives marked variability in Pacific and monsoon Asia, the Hadley circulation remains the dominant, consistent determinant of tropical rainfall.

Levels marked using Generic Level Descriptors for 20m H2 essays

6 'Weathering is the main factor influencing mass movements in tropical climates.'

Evaluate this statement. [20]

Possible Approaches

Candidates could evaluate the extent to which weathering influences mass movements by considering both its preparatory role and how it compares with other natural and human factors. They could argue that in humid tropical climates, deep chemical weathering (e.g., hydrolysis, oxidation) weakens rock structures and forms thick regolith, predisposing slopes to mass movement. In contrast, in arid/semi-arid tropical areas, weathering is less intense, and mass movements are often triggered by sudden rainfall or human activities.

Candidates could also examine the importance of rainfall intensity, slope gradient, and vegetation cover, which directly affect slope stability. Human activities like deforestation and slope undercutting may exacerbate or independently trigger mass movements. Stronger responses may adopt a systems approach, recognising that weathering interacts with climatic and anthropogenic factors to influence slope failure. They may also apply geographical concepts such as *scale* and *place*, showing how influences vary across tropical regions. Candidates are expected to make a reasoned judgment on whether weathering is the main or one of several key influences on mass movements in the tropics, supported by case studies and examples.

Levels marked using Generic Level Descriptors for 20m H2 essays

Levels	Marks	Generic Level Descriptors for H2 Essays
5	18–20	Evaluation is consistently analytical and coherent. Response is well-supported by relevant material, including the effective use of examples. Response features accurate geographical knowledge and reflects good understanding of the subject content relevant to the question.
4	14–17	Evaluation is analytical and coherent. Response is mostly well-supported by relevant material, including the appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
3	10–13	Evaluation is broadly analytical and generally coherent. Response is moderately well-supported by relevant material, including some appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
2	6–9	Response is largely descriptive with limited analysis and evaluation. Response is partly coherent and may lack clarity in parts. Response is poorly supported by relevant materials, including the limited use of examples. Response features inaccurate geographical knowledge and poor understanding of the subject content relevant to the question.
1	1–5	Response is descriptive with no analysis or evaluation. Response is fragmented and lacks clarity. Response consists of unsupported assertions. Response features largely inaccurate geographical knowledge and a lack of understanding of the subject content relevant to the question.
0	0	No creditworthy response